

CUNDERDIN, AustraliaWMO: **956250**Lat: **31.6219S**Lon: **117.2217E**Elev: **713**StdP: **14.32**Time Zone: **8.00 (E08)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.3	35.3	21.0	16.2	76.8	26.1	20.6	65.7	27.5	61.0	25.0	61.1	3.0	70	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	102.8	67.4	99.3	66.7	96.0	65.9	72.3	89.7	70.2	89.4	68.6	88.4	12.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.7	104.3	77.4	64.9	94.5	75.6	62.5	87.0	74.2	36.4	90.0	34.6	89.2	33.2	88.0	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.2	19.8	29.6	109.3	1.5	3.0	28.5	111.5	27.6	113.3	26.8	115.0	25.7	117.2		
(5)			28.7	75.3	1.8	2.8	27.5	77.3	26.5	78.9	25.5	80.5	24.2	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	78.1	77.5	73.3	66.4	58.8	53.7	51.5	51.9	54.7	62.2	70.2	74.7		
	DBStd	11.31	6.20	6.02	6.43	5.93	5.10	4.30	4.29	4.28	5.05	6.25	6.56	6.85		
	HDD50	88	0	0	0	0	1	14	35	27	11	0	0	0		
	HDD65	1903	1	1	8	53	203	340	419	405	312	131	24	6		
	CDD50	5323	872	770	724	492	273	124	81	87	152	378	605	765		
	CDD65	1661	408	351	267	94	10	0	0	0	2	44	179	306		
	CDH74	21033	5279	4318	3011	934	124	2	0	9	126	819	2383	4028		
	CDH80	10885	3039	2377	1458	307	21	0	0	1	29	291	1162	2200		
(14) Wind	WSAvg	9.7	12.0	10.9	11.0	9.0	7.9	8.2	8.2	8.0	8.8	9.9	11.2	11.7		
(15) Precipitation	PrecAvg	14.6	0.6	0.7	0.7	1.0	1.9	2.6	2.4	1.9	1.2	0.7	0.6	0.4		
	PrecMax	23.7	5.3	3.7	4.5	3.4	5.0	7.5	5.4	5.1	3.1	2.6	3.5	1.7		
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.7	0.3	0.0	0.0	0.0		
	PrecStd	3.3	1.1	0.9	0.9	0.8	1.3	1.3	1.1	0.8	0.6	0.5	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.6	105.7	101.2	92.7	82.6	73.1	69.5	74.2	83.8	94.1	101.4	106.4		
		MCWB	68.3	69.4	66.3	63.5	58.2	55.1	54.9	57.4	59.4	62.3	64.2	66.2		
	2%	DB	102.7	100.9	97.3	87.3	77.4	69.6	65.8	68.9	77.4	87.8	95.5	101.0		
		MCWB	67.8	68.4	66.3	62.2	58.0	54.8	54.8	56.4	57.8	61.5	63.3	65.8		
	5%	DB	98.6	97.0	93.0	83.3	73.5	66.3	63.0	65.3	72.0	82.3	91.5	96.4		
		MCWB	67.3	67.2	66.1	61.7	57.2	54.5	54.4	55.9	57.0	59.8	62.0	64.9		
	10%	DB	94.5	93.2	88.1	79.4	70.0	63.4	60.7	62.3	67.9	78.1	86.9	91.9		
		MCWB	66.2	66.6	64.6	60.7	57.0	54.2	53.7	54.7	55.8	58.4	61.4	63.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.8	75.5	71.5	68.2	64.5	61.3	60.0	61.2	62.5	65.9	68.0	70.9		
		MCDB	90.3	91.4	88.0	80.1	70.7	66.5	64.4	67.7	76.8	84.4	90.7	93.1		
	2%	WB	72.6	72.6	69.4	66.0	62.1	59.0	57.5	59.1	60.3	63.4	66.1	68.6		
		MCDB	88.9	90.2	85.8	78.9	69.5	64.4	61.8	65.0	71.9	81.3	87.7	92.0		
	5%	WB	70.5	70.2	67.9	64.4	60.5	57.3	56.0	57.3	58.4	61.5	64.4	66.7		
		MCDB	89.6	90.3	85.7	77.6	68.5	62.8	60.7	63.3	68.4	77.9	84.5	89.9		
	10%	WB	68.4	68.4	66.4	62.8	58.9	55.6	54.5	55.4	56.6	59.8	62.7	64.9		
		MCDB	89.6	87.8	83.2	75.2	66.7	60.9	59.4	61.2	66.1	74.8	82.9	86.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	28.1	27.1	25.5	23.8	22.2	20.4	19.5	21.8	26.4	29.2	28.5	28.9		
		MCDBR	34.6	33.1	31.6	29.2	28.0	25.5	22.4	27.1	34.6	36.3	35.8	35.2		
		MCWBR	11.3	10.6	11.2	11.6	13.5	14.2	13.8	16.4	17.7	14.9	12.2	12.0		
	5% WB	MCDBR	27.5	28.1	25.8	23.4	19.6	18.7	17.3	21.5	28.1	30.6	30.5	31.1		
		MCWBR	10.2	10.2	10.1	10.5	11.2	12.5	11.9	14.4	15.7	13.5	11.0	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.335	0.319	0.308	0.293	0.285	0.285	0.291	0.313	0.319	0.320	0.324		
	taud		2.505	2.514	2.544	2.557	2.579	2.590	2.590	2.572	2.484	2.491	2.507	2.503		
	Ebn at Noon		319	312	307	294	284	278	284	296	304	314	321	323		
	Edh at Noon		36	35	31	28	24	23	23	27	33	35	36	36		
(44) All-Sky Solar Radiation	RadAvg		2588	2337	1947	1445	1099	909	960	1236	1639	2104	2477	2670		
	RadStd		138	104	106	84	78	57	49	53	77	117	114	127		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+2.41	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page